

REQUIRED DELIVERABLE

Evaluation Report

Metrics, test protocol and current internal results for the COMPASSIQ prototype.

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Prepared for GitHub repository submission

1. Evaluation Scope

This report documents how COMPASSIQ is evaluated across the three mandatory modules: adaptive maturity diagnosis, explainable scoring, and RAG-grounded roadmap generation. The objective is to show that the system produces measurable, traceable and actionable outputs rather than generic chatbot responses.

Current results are based on a small internal test set suitable for a hackathon prototype. They should be expanded with real entrepreneurs and ecosystem experts during the pilot phase.

2. Test Set Design

Test asset	Size	Purpose
Labelled founder profiles	20 profiles	Evaluate maturity classification across Ideation, Market Validation, Structuration, Fundraising, Launch Planning and Growth.
Perception-gap cases	3 cases	Check whether the system flags divergence between self-assessed stage and evidence-based diagnosis.
Scoring stability prompts	10 repeated profile runs	Measure whether score outputs remain consistent for equivalent profile evidence.
RAG roadmap scenarios	25 retrieval checks	Evaluate whether retrieved resources match stage, sector and detected blocker.
Roadmap coherence reviews	5 complete scenarios	Check whether recommendations are ordered, sourced and actionable.

Profile dimensions used in the tests: sector, legal form, team composition, revenue status, validation evidence, business model clarity, geographic scope, prior accompaniment history, self-assessed maturity and supporting evidence.

3. Evaluation Metrics

Metric	Formula / method	Target behavior
Maturity classification accuracy	Correct stage predictions / total labelled profiles	The diagnostic engine assigns the maturity stage supported by evidence.
Perception-gap detection rate	Correctly flagged divergence cases / total divergence cases	The system detects founders who overestimate or underestimate readiness.
Scoring consistency	Average normalized stability across repeated equivalent profiles	Scores remain stable when evidence is equivalent and change when evidence changes.
Top-3 RAG relevance	Relevant resource retrieved in top 3 / total retrieval checks	Roadmap resources are aligned with sector, stage and blocker.
Roadmap coherence score	Expert checklist average on 1-5 scale	Actions are ordered, sourced, realistic and connected to diagnosis/scoring.

4. Current Internal Results

Metric	Result	Interpretation
Maturity classification accuracy	85% (17/20)	Most profiles are correctly assigned; remaining errors are boundary cases between Market Validation and Structuration.
Perception-gap detection	3/3 cases	All designed divergence cases were flagged with explicit evidence.

Metric	Result	Interpretation
Scoring consistency	0.82	Scores are stable across equivalent inputs; calibration with experts remains a next step.
Top-3 RAG relevance	88% (22/25)	Most scenarios retrieve a relevant Tunisian resource within the top three results.
Roadmap coherence	4.4/5	Roadmaps are generally actionable, ordered and source-grounded.

Note: these are hackathon-stage internal results. Before production, the test set should include real founders, support-program officers and sector experts.

5. Error Analysis

Observed issue	Example	Mitigation
Boundary-stage ambiguity	A project with interviews but no paid pilot may sit between Market Validation and Structuration.	Surface confidence level and ask follow-up evidence questions.
Sparse evidence	Founder claims traction without numbers or customer proof.	Flag uncertainty and request validation artefacts before assigning higher maturity.
Resource overlap	A financing resource may apply to several stages.	Rank by stage fit, sector fit, blocker fit and source confidence.
Synthetic test bias	Internal profiles may not reflect all real founder behaviors.	Run pilot with real entrepreneurs and mentors in Tunisia.

6. Sample Evaluation Cases

Scenario	Expected output	Observed behavior
AgriTech founder self-assesses as Fundraising but lacks customer validation.	Classify lower than Fundraising; flag perception gap; recommend validation and AgriTech support.	Classified as Structuration; roadmap cites kb_024 Cap Bon AgriTech Hub, kb_025 APIA and kb_036 CTAB.
FinTech founder with prototype but unclear regulatory readiness.	Detect regulatory blocker; retrieve relevant administrative / sandbox resources.	Roadmap prioritizes regulatory review and BCT sandbox-oriented resources.
Growth-stage SME seeking export support.	Classify Growth / Launch Planning; retrieve export and standards resources.	Retrieved CEPEX and standards-oriented resources where applicable.

7. Next Evaluation Steps

- Expand labelled profiles from 20 to at least 100 profiles across sectors and maturity stages.
- Use inter-rater agreement with mentors/investors to calibrate diagnostic labels and scoring weights.
- Evaluate Arabic/French intake separately to detect language-related performance gaps.
- Track click-through or completion of roadmap actions in the “Mon Parcours” view.
- Re-evaluate RAG relevance after every knowledge base update.